

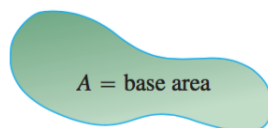
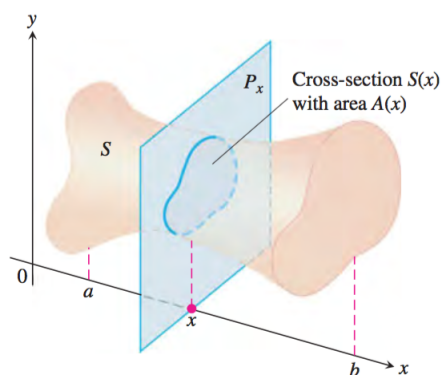
6.2 Volumes (Using Cross-Sections)

How do we calculate the volume of an object?

Slicing by Parallel Planes

Definition

We define a _____ of a solid S as the plane region formed by intersecting S with a plane.



Plane region whose
area we know



Cylindrical solid based on region
Volume = base area $\times$ height = Ah

Suppose that we partition $[a, b]$ into subintervals

$$a = x_0 < x_1 < x_2 < x_3 < x_4 < \dots < x_n = b,$$

where each subinterval has width Δx_k . What is the volume of one slice of our solid?

How would we use this to calculate the volume of the entire solid?

Definition

If the _____ of a solid S at each point x in a region $[a, b]$ is a region $S(x)$ of Area $A(x)$, where $A(x)$ is a _____ function of x , then

$$V(s) =$$

Calculating the Volume of a Solid

1. Sketch a picture of the solid and _____.
2. Find a formula for _____.
3. Find the _____.
4. Integrate _____ to find the volume.

Let's write out some typical cross-sectional area formulae:

Example 1

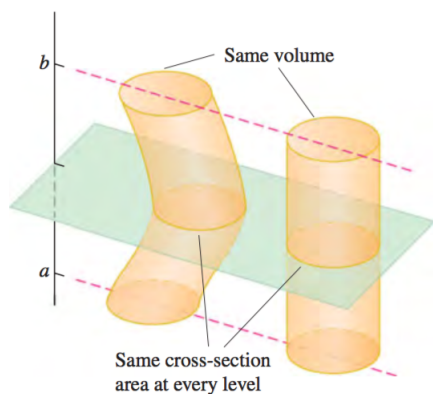
A cone with height 5 cm has a circular base with radius 5 cm.
Find the volume of the cone.

Example 2

Set up an integral to find the volume of the solid that lies between planes perpendicular to the x -axis at $x = 0$ and $x = 4$. The cross-sections, perpendicular to the x -axis between these planes, are squares with bases in the xy -plane that run from one side of the parabola $x = y^2$ to the other.

Cavalieri's Principle

Solids with equal _____ and identical
_____ at each height have the same volume.



Why is this important?

When setting up an integral that represents the volume of a solid found by slicing, how do we know whether the variable of integration should be x or y ?

Extra Examples

Try these examples on your own and compare them to the videos.

1. Set up an integral to find the volume of the given solid, S . The base of S is the triangular region with vertices $(0, 0)$, $(1, 0)$, and $(0, 1)$. Cross-sections perpendicular to the y -axis are semi-circles with diameters running across the base of the solid.

2. Set up an integral to find the volume of the given solid. The base of a solid is the region bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{2}$. The cross-sections perpendicular to the x -axis are

(a) isosceles triangles of height 6

(b) equilateral triangles